Devanshu Bhargava

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github.com/devanshubhargava Portfolio

Summary

Full Stack Software Engineer with 2+ years of experience building scalable **Java** and **Spring Boot** microservices and event-driven systems on **Apache Kafka**, Redis and PostgreSQL, engineered for fault tolerance with DLQs, circuit breakers, outbox and idempotency. Pairs strong backend engineering with hands-on **Generative AI**: hybrid RAG and GraphRAG pipelines, **LangGraph** multi-agent systems, retrieval evaluation and cost-aware LLM routing on Amazon Bedrock and Ollama. Ships end to end with React/Next.js frontends, Docker and AWS.

Experience

Infinite Computer Solutions **Oct 2024 – Present**
Associate Software Engineer *Noida*

- **High-Throughput Telemetry Platform:** Build and maintain **Spring Boot** services for an enterprise real-time **fleet tracking platform** serving **50K+ active drivers**, ingesting **15,000+ GPS events/sec** in a **10+ engineer** team.
- **Reliable Event Streaming:** Engineered **Apache Kafka consumers** with manual offset commits, graceful shutdown hooks and rebalance listeners, ensuring **no GPS event is lost** across crashes and partition rebalances.
- **Fault Isolation:** Built a **batch-bisection recovery** flow and a **Dead Letter Queue (DLQ)** tagged by error class, isolating malformed pings without failing valid rows and cutting triage and replay time by **80%**.
- **Throughput Optimization:** Boosted **PostgreSQL** write throughput **12x** by replacing single-row inserts with 1,000-record Kafka batches and **COPY**-based bulk loads with idempotent upserts, making replays duplicate-safe.
- **Real-Time Fan-Out:** Integrated **Redis Pub/Sub** with Lua-based stale-update guards and bounded-queue **WebSocket** gateways, streaming live vehicle positions to dispatchers across instances without slow-consumer memory growth.

Projects

SENTINEL — Event-Driven Fraud Detection Gateway | *Java, Spring Boot, Kafka, Redis, Python, FastAPI*

- **Async Payment Processing:** Event-driven gateway on **Kafka** that moves fraud analysis off the request path, cutting API latency from **2s+ to 50ms**.
- **Resilient Delivery: Resilience4j** circuit breaker, **Transactional Outbox** and DLQ for reliable fraud-event delivery, with **idempotency keys** preventing duplicate checks.
- **Rate Limiting & ML Inference:** Distributed **Redis token-bucket** limiter shared across API instances, plus a **FastAPI** ML inference service for scalable fraud scoring.

CORTEX — Multi-Tenant GraphRAG Research Platform | *Python, LangGraph, pgvector, Neo4j, Bedrock*

- **Hybrid Retrieval:** Postgres full-text + **pgvector** HNSW, Reciprocal Rank Fusion, cross-encoder reranking and **Neo4j GraphRAG**; recall@k and MRR evaluated nightly in CI on a hand-labelled golden set.
- **Multi-Agent Orchestration: LangGraph** supervisor over Researcher, Analyst and Coder agents with loop guard and token budgets; cost-aware Ollama/Bedrock routing; sandboxed code execution.
- **Multi-Tenant Platform:** Tenant isolation with PostgreSQL **Row-Level Security**; SSE streaming from **Celery** workers to a Next.js UI with citations.

Technical Skills

Languages: Java 21, Python, TypeScript, JavaScript, SQL
Backend & Distributed Systems: Spring Boot, FastAPI, gRPC, REST, Microservices, Kafka, Redis, Celery, Resilience4j
Databases: PostgreSQL, TimescaleDB, pgvector, Neo4j
Generative AI & ML: RAG, LangGraph, LangChain, Amazon Bedrock, Ollama, scikit-learn, MLflow
Cloud & DevOps: AWS (ECS, RDS, ElastiCache, S3, Cognito), Terraform, Docker, Kubernetes, GitHub Actions
Frontend & Testing: React, Next.js, Testcontainers, JUnit 5, pytest, k6, OpenTelemetry, Prometheus, Grafana

Education

Rajiv Gandhi Proudyogiki Vishwavidyalaya **June 2020 – June 2024**
B.Tech, Computer Science Engineering | CGPA: 7.62 *Madhya Pradesh*

Achievements

Google Data Analytics & Advanced Data Analytics Professional Certificates — Verify Verify
Performer of the Month among 200+ employees — Verify